

Question 1

$$f(x) = \begin{cases} x^2 - 3x - 4 & \text{if } x < 0 \\ -\cos x - 4 & \text{if } x > 0 \end{cases}$$

$$(i) \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (x^2 - 3x - 4) = -4$$

$$(ii) \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (-\cos x - 4) = -5$$

$$(iii) \lim_{x \rightarrow 0} f(x) \text{ does not exist because } \lim_{x \rightarrow 0^-} f(x) \neq \lim_{x \rightarrow 0^+} f(x)$$

Question 2

$$f(x) = \frac{10 - x^2}{x - 5}$$

$$(i) \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{10 - x^2}{x - 5} = \frac{10}{-5} = -2$$

$$(ii) \lim_{x \rightarrow 5^+} f(x) = \lim_{x \rightarrow 5^+} \frac{10 - x^2}{x - 5} = -\infty$$

(because if $x \rightarrow 5^+$ then $10 - x^2$ is negative and $x - 5$ is positive.)

$$(iii) \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{10/x - x}{1 - 5/x} = -\infty$$

Question 3

$$f(x) = \frac{x - 8}{x^2 - 64} = \frac{x - 8}{(x - 8)(x + 8)} = \frac{1}{x + 8} \text{ if } x \neq 8$$

$$(i) \lim_{x \rightarrow 8^-} f(x) = \lim_{x \rightarrow 8^-} \frac{1}{(x + 8)} = \frac{1}{16}$$

$$(ii) \lim_{x \rightarrow 8^+} f(x) = \lim_{x \rightarrow 8^+} \frac{1}{(x + 8)} = \frac{1}{16}$$

$$(iii) \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{x - 8}{x^2 - 64} = \lim_{x \rightarrow \infty} \frac{1/x - 8/x^2}{1 - 64/x^2} = \frac{0}{1} = 0$$

OR

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{1}{x + 8} = 0$$

Question 4

$$f(x) = \frac{x - 10}{x^2 + 2x + 1} = \frac{x - 10}{(x + 1)^2}$$

$$(i) \lim_{x \rightarrow -1^-} \frac{x - 10}{(x + 1)^2} = -\infty$$

(because if $x \rightarrow -1^-$ then $x - 10$ is negative and $(x + 1)^2$ is positive.)

$$(ii) \lim_{x \rightarrow -1^+} \frac{x - 10}{(x + 1)^2} = -\infty$$

(because if $x \rightarrow -1^+$ then $x - 10$ is negative and $(x + 1)^2$ is positive.)

$$(iii) \lim_{x \rightarrow -1} f(x) = -\infty$$

Question 5

(i) $\lim_{x \rightarrow 0} \sin x = \sin 0 = 0$

(ii) $\lim_{x \rightarrow 0} \cos x = \cos 0 = 1$

(iii) $\lim_{x \rightarrow 0^-} \operatorname{cosec} x = \lim_{x \rightarrow 0^-} \frac{1}{\sin x} = -\infty$

(because if $x \rightarrow 0^-$ then 1 is positive and $\sin x$ is negative.)

(iv) $\lim_{x \rightarrow 0^+} \cot x = \lim_{x \rightarrow 0^+} \frac{\cos x}{\sin x} = \infty$

(because if $x \rightarrow 0^+$ then $\cos x$ is positive and $\sin x$ is positive.)

(v) $\lim_{x \rightarrow \pi/2} \sin x = \sin \pi/2 = 1$

(vi) $\lim_{x \rightarrow \pi/2} \cos x = \cos \pi/2 = 0$

(vii) $\lim_{x \rightarrow \pi/2^+} \sec x = \lim_{x \rightarrow \pi/2^+} \frac{1}{\cos x} = -\infty$

(because if $x \rightarrow \pi/2^+$ then 1 is positive and $\cos x$ is negative.)

(viii) $\lim_{x \rightarrow \pi/2^+} \tan x = \lim_{x \rightarrow \pi/2^+} \frac{\sin x}{\cos x} = -\infty$

(because if $x \rightarrow \pi/2^+$ then $\sin x$ is positive and $\cos x$ is negative.)

Question 6

(i) $\lim_{x \rightarrow \pi} \frac{\sin x}{x} = \frac{\sin \pi}{\pi} = \frac{0}{\pi} = 0$

(ii) $\lim_{x \rightarrow \pi} \frac{\sin(x^2 - \pi^2)}{x - \pi} = \lim_{x \rightarrow \pi} \left[\frac{\sin[(x - \pi)(x + \pi)]}{x - \pi} \times \frac{x + \pi}{x + \pi} \right]$
 $= \lim_{x \rightarrow \pi} \frac{\sin(x - \pi)(x + \pi)}{(x - \pi)(x + \pi)} \times \lim_{x \rightarrow \pi} (x + \pi) = 1 \times 2\pi = 2\pi$

(iii) $\lim_{x \rightarrow \pi} \frac{\sin(x - \pi)}{x^2 - \pi^2} = \lim_{x \rightarrow \pi} \frac{\sin(x - \pi)}{(x - \pi)(x + \pi)}$
 $= \lim_{x \rightarrow \pi} \frac{\sin(x - \pi)}{(x - \pi)} = \times \lim_{x \rightarrow \pi} \frac{1}{(x + \pi)} = 1 \times \frac{1}{2\pi} = \frac{1}{2\pi}$

(iv) $\lim_{x \rightarrow 0} \frac{\tan 3x}{\sin 6x} = \lim_{x \rightarrow 0} \left[\frac{\tan 3x}{\sin 6x} \times \frac{3x}{3x} \times \frac{6x}{6x} \right]$
 $= \lim_{x \rightarrow 0} \left[\frac{\tan 3x}{3x} \times \frac{6x}{\sin 6x} \times \frac{3x}{6x} \right] = \lim_{x \rightarrow 0} \frac{\tan 3x}{3x} \times \lim_{x \rightarrow 0} \frac{6x}{\sin 6x} \times \lim_{x \rightarrow 0} \frac{3}{6} = 0.5$

Question 7

(i) $\lim_{x \rightarrow \infty} \frac{1 - 4x^2}{2x^2 + 4x - 6} = \lim_{x \rightarrow \infty} \frac{1/x^2 - 4}{2 + 4/x - 6/x^2} = -2$

(ii) $\lim_{x \rightarrow -\infty} \frac{1 - 4x^3}{2x^2 + 4x - 6} = \lim_{x \rightarrow -\infty} \frac{1/x^2 - 4x}{2 + 4/x - 6/x^2} = \infty$

(iii) $\lim_{x \rightarrow \infty} \frac{1 - 4x^5}{2x^6 + 4x - 6} = \lim_{x \rightarrow \infty} \frac{1/x^6 - 4/x}{2 + 4/x^5 - 6/x^6} = 0$

Question 8

$$\begin{aligned}
\text{(i)} \quad \lim_{x \rightarrow 1^+} \frac{\sqrt{x-1}}{\sqrt{x}-1} &= \lim_{x \rightarrow 1^+} \left[\frac{\sqrt{x-1}}{\sqrt{x}-1} \times \frac{\sqrt{x}+1}{\sqrt{x}+1} \right] \\
&= \lim_{x \rightarrow 1^+} \frac{\sqrt{x-1}(\sqrt{x}+1)}{x-1} = \lim_{x \rightarrow 1^+} \frac{\sqrt{x}+1}{\sqrt{x}-1} = \infty \\
\text{(ii)} \quad \lim_{x \rightarrow 4} \frac{x-4}{\sqrt{x+21}-5} &= \lim_{x \rightarrow 4} \left[\frac{x-4}{\sqrt{x+21}-5} \times \frac{\sqrt{x+21}+5}{\sqrt{x+21}+5} \right] \\
&= \lim_{x \rightarrow 4} \frac{(x-4)(\sqrt{x+21}+5)}{x+21-25} = \lim_{x \rightarrow 4} \frac{(x-4)(\sqrt{x+21}+5)}{x-4} \\
&= \lim_{x \rightarrow 4} (\sqrt{x+21}+5) = 10 \\
\text{(iii)} \quad \lim_{x \rightarrow 3} \frac{3-x}{\frac{1}{x}-\frac{1}{3}} &= \lim_{x \rightarrow 3} \frac{3-x}{\frac{3-x}{3x}} = \lim_{x \rightarrow 3} \left[\frac{3-x}{1} \times \frac{3x}{3-x} \right] = \lim_{x \rightarrow 3} 3x = 9
\end{aligned}$$

Question 9

$$\begin{aligned}
\text{(i)} \quad \lim_{x \rightarrow \infty} e^{-3x} &= 0 \\
\text{(ii)} \quad \lim_{x \rightarrow -\infty} e^{3x} &= 0 \\
\text{(iii)} \quad \lim_{x \rightarrow \infty} e^{5x} &= \infty
\end{aligned}$$

Question 10

$$\begin{aligned}
\text{(i)} \quad \lim_{x \rightarrow 1} \ln x &= \ln 1 = 0 \\
\text{(ii)} \quad \lim_{x \rightarrow \infty} \ln x &= \infty \\
\text{(iii)} \quad \lim_{x \rightarrow 0^-} \ln x &\text{ DNE (Do not exist)} \\
\text{(iv)} \quad \lim_{x \rightarrow 2^+} \ln(x-2) &= -\infty
\end{aligned}$$

Question 11

$$\begin{aligned}
\text{(i)} \quad \lim_{x \rightarrow 0} \arctan x &= \arctan 0 = 0 \\
\text{(ii)} \quad \lim_{x \rightarrow \infty} \arctan(2x-3) &= \frac{\pi}{2} \\
\text{(iii)} \quad \lim_{x \rightarrow -\infty} \arctan(-2x+3) &= -\frac{\pi}{2}
\end{aligned}$$

Question 12

$$\begin{aligned}
\text{(i)} \quad \lim_{x \rightarrow 1} \frac{|x|-1}{x-1} &= \lim_{x \rightarrow 1} \frac{x-1}{x-1} \quad \text{because if } x \rightarrow 1 \text{ then } x > 0 \text{ and } |x| = x \\
&= \lim_{x \rightarrow 1} 1 = 1 \\
\text{(ii)} \quad \lim_{x \rightarrow 0} \frac{|x|-1}{x-1} &= \frac{-1}{-1} \quad (\text{using direct substitution}) = 1 \\
\text{(iii)} \quad \lim_{x \rightarrow 1^-} \frac{|x-1|}{x-1} &= \lim_{x \rightarrow 1^-} \frac{-(x-1)}{x-1} = \lim_{x \rightarrow 1^-} -1 = -1
\end{aligned}$$

Question 13

$$(i) \lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$$

$$(ii) \lim_{x \rightarrow 0} \sin\left(\frac{1}{x^2}\right) \text{ DNE}$$

$$(iii) \lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x^2}\right) = 0 \text{ using the squeezing theorem:}$$

$$-1 \leq \sin\left(\frac{1}{x^2}\right) \leq 1 \text{ for } x \neq 0$$

$$\Rightarrow -x^2 \leq x^2 \sin\left(\frac{1}{x^2}\right) \leq x^2$$

$$\text{and as } \lim_{x \rightarrow 0} (-x^2) = \lim_{x \rightarrow 0} x^2 = 0 \text{ it follows that } \lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x^2}\right) = 0$$

Question 14

$$(i) \lim_{x \rightarrow -\infty} \frac{e^{-x}}{e^x - 1} = -\infty \text{ (Because if } x \rightarrow -\infty \text{ then } e^{-x} \rightarrow \infty \text{ and } e^x - 1 \rightarrow -1)$$

$$(ii) \lim_{x \rightarrow -\infty} \frac{e^x}{e^{-x} - 1} = 0 \text{ (Because if } x \rightarrow -\infty \text{ then } e^x \rightarrow 0 \text{ and } e^{-x} - 1 \rightarrow \infty)$$

$$(iii) \lim_{x \rightarrow \infty} \frac{e^{2x}}{e^{3x} - 1} = \lim_{x \rightarrow \infty} \frac{e^{2x}/e^{3x}}{1 - 1/e^{3x}} = \lim_{x \rightarrow \infty} \frac{1/e^{2x}}{1 - 1/e^{3x}} = \frac{0}{1} = 0$$